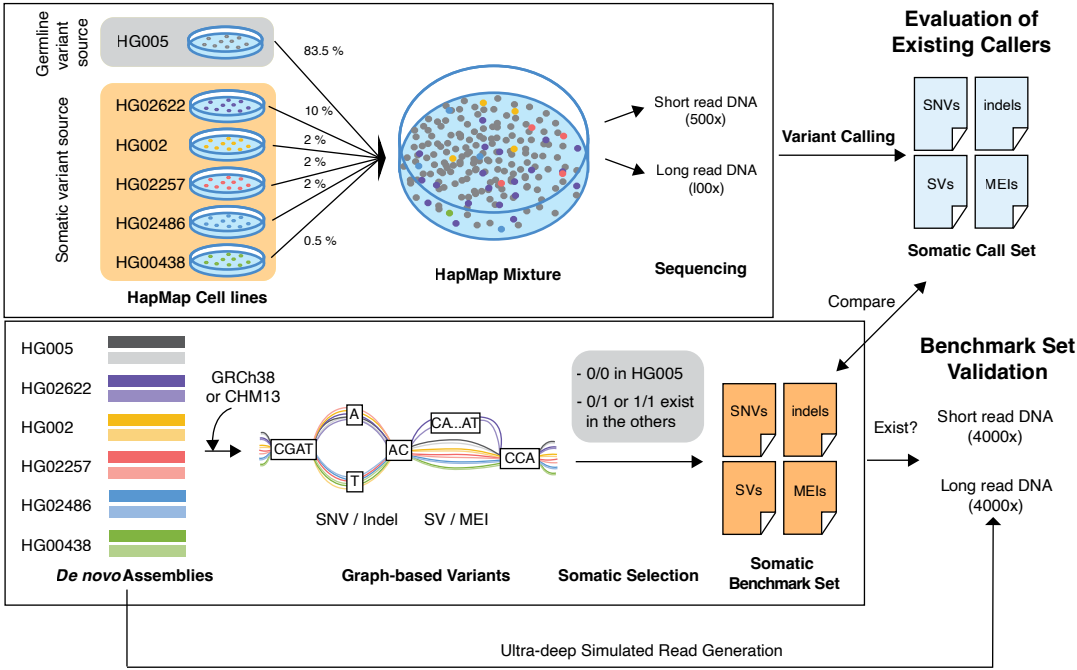
Benchmark Sample
Germline variant source
HG005
83.5 %
Somatic variant source
HG02622
10 %
HG002
2 %
HG02257
2 %
HG02486
0.5 %
HG00438
HapMap Cell lines
HapMap Mixture
Short read DNA (500x)
Long read DNA (l00x)
Sequencing
Variant Calling
SNVs
indels
SVs
MEIs
Somatic Call Set
Evaluation of Existing Callers
Compare
Benchmark Set Validation
Benchmark Variant
HG005
HG02622
HG002
HG02257
HG02486
HG00438
De novo Assemblies
GRCh38 or CHM13
CGAT
A
T
AC
CA...AT
CCA
SNV / Indel
SV / MEI
Graph-based Variants
- 0/0 in HG005
- 0/1 or 1/1 exist in the others
Somatic Selection
SNVs
indels
SVs
MEIs
Somatic Benchmark Set
Exist?
Short read DNA (4000x)
Long read DNA (4000x)
Ultra-deep Simulated Read Generation